

Silvia Sapura

silvia.sapura@gmail.com | linkedin.com/in/silvia-sapura | github.com/SilviaSapura

SUMMARY

I am currently pursuing a PhD in Statistics at the University of Oxford, where my research focuses on **Reinforcement Learning**, **Meta-Learning**, and **Large Language Models**. With a strong foundation in applied machine learning from my tenure at Meta, I aim to advance the field through innovative research and practical applications.

EDUCATION

University of Oxford <i>PhD in Statistics, supervised by Yee Whye Teh and Jakob Foerster</i>	Oct 2022 – Ongoing Oxford, UK
Imperial College London <i>Master of Engineering in Computer Science (incl. Bachelor)</i> • First Class, Dean's List (top 10% of the year)	Oct 2015 – June 2019 London, UK

WORK EXPERIENCE

Machine Learning Research Intern <i>Apple</i> • Developed a novel framework combining LLMs and evolutionary strategies to improve Inverse Reinforcement Learning techniques. Authored a research paper on the findings accepted to ICLR 2026.	Apr 2025 – Sept 2025 London, UK
Teaching Assistant <i>University of Oxford, Imperial College London</i> • Courses: Applied Probability, Statistical Machine Learning, Deep Learning	Oct 2022 – Apr 2025 London, UK
Machine Learning Research Intern <i>SamayaAI</i> • Improved Information Retrieval methods using Large Language Models.	Jun 2024 – Feb 2025 London, UK
Software Engineer <i>Meta</i> • Specialized in applied machine learning to enhance Meta's developer infrastructure. • Worked on identifying relevant data, building data pipelines, data cleaning and model optimisation. • Co-authored and published multiple papers on novel approaches to improving reliability, integrity, and privacy in developer systems.	Aug 2019 – Oct 2022 London, UK
Software Engineering Intern <i>Meta</i>	Apr 2018 – Sep 2018 London, UK
Software Engineering Intern <i>Redgate Software</i> • Implemented a Machine Learning Anomaly Detection model to detect anomalies in metrics such as CPU and memory usage and presented the feature in a company-wide meeting.	Jun 2017 – Sep 2017 Cambridge, UK
Teacher <i>Turing Lab</i> • Taught kids aged 8-16 the basics of coding using Scratch, JavaScript and Python.	Jan 2016 – Jun 2019 London, UK

RESEARCH

- Sapora S., Hjelm D., Toshev A., Mazouze B., GRACE: A Language Model Framework for Explainable Inverse Reinforcement Learning, preprint: arXiv:2510.02180, *14th International Conference on Learning Representations (ICLR)*, 2026
- Cook J.*, Sapora S.*, Ahmadian A., Khan A., Rocktäschel T., Foerster J., Ruis L., Programming by Backprop: LLMs Acquire Reusable Algorithmic Abstractions During Code Training, arXiv preprint: arXiv:2506.18777, *14th International Conference on Learning Representations (ICLR)*, 2026, **equal contributions*.
- Alfano C.*, Sapora S.*, Foerster J., Rebeschini P., Teh Y.W., Meta-Learning Objectives for Preference Optimization, 39th Conference on Neural Information Processing Systems (NeurIPS 2025), **equal contributions*.
- Sapora S., Baziotis C., Dessi R., Petroni P., Bevilacqua M. QCR: Quantised Codebooks for Retrieval, *under review*.
- Sapora S., Swamy G., Lu C., Teh Y.W., Foerster J. EvIL: Evolution Strategies for Generalisable Imitation Learning, *Proceedings of the 41st International Conference on Machine Learning, Vienna, Austria. PMLR 235*, 2024.
- DeMoss B., Sapora S., Foerster J. Hawes N., Posner I. The Complexity Dynamics of Grokking, Physica D: Nonlinear Phenomena
- Alfano C., Towers S., Sapora S., Lu C., Rebeschini P. Meta-learning the mirror map in policy mirror descent, *Proceedings of the 13th International Conference on Learning Representations*, 2025
- Nagy P., Frey S., Sapora S., Li K., Calinescu A., Zohren S., Foerster J. Generative AI for End-to-End Limit Order Book Modelling: A Token-Level Autoregressive Generative Model of Message Flow Using a Deep State Space Network, *Proceedings of the Fourth ACM International Conference on AI in Finance*, 91-99, 2023
- Frey S., Li K., Nagy P., Sapora S., Zohren S., Foerster J., Calinescu A. JAX-LOB: A GPU-Accelerated limit order book simulator to unlock large scale reinforcement learning for trading, *Proceedings of the Fourth ACM International Conference on AI in Finance*, 583-591, 2023
- Ahlgren J. et al. Behavioural and structural imitation models in Facebook's WW simulation system, *9th International Workshop on Realizing Artificial Intelligence Synergies in Software Engineering (RAISE 2021)*
- Bojarczuk K. et al. Measurement challenges for cyber digital twins: Experiences from the deployment of Facebook's ww simulation system, *Proceedings of the 15th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*, 2021
- Ahlgren J. et al. Facebook's cyber-cyber and cyber-physical digital twins, *Proceedings of the 25th International Conference on Evaluation and Assessment in Software Engineering*, 2021
- Ahlgren J. et al. Testing web enabled simulation at scale using metamorphic testing, *IEEE/ACM 43rd International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)*, 2021
- Ahlgren J. et al. Ownership at large: Open problems and challenges in ownership management, *Proceedings of the 28th International Conference on Program Comprehension*, 2020
- Ahlgren J. et al. WES: Agent-based user interaction simulation on real infrastructure, *Proceedings of the IEEE/ACM 42nd International Conference on Software Engineering Workshops*, 2020
- Sapora S. Grasp Quality Deep Neural Networks for Robotic Object Grasping, Master's Thesis, 2019
- Sapora S., Lazarescu B., Lolov C. Absit invidia verbo: Comparing deep learning methods for offensive language, arXiv preprint: arXiv:1903.05929, 2019

SELECTED TALKS

- **Behavioural and structural imitation models in Facebook's WW simulation system**, International AI for Agent-Based Modelling Community Meeting, April 2022
- **AI Techniques for Simulating Behaviour using Facebook's WW Social Media Simulation System**, 9th workshop on Realising Artificial Intelligence Synergies with Software Engineering, November 2021
- **Ownership at Large - Open Problems and Challenges in Ownership Management**, 28th International Conference on Program Comprehension, 2021 [link]
- **WES: Agent-based User Interaction Simulation on Real infrastructure**, Facebook, London Tech Talks, for an audience of 500+, December 2019 and London Tech Summit, November 2019
- **Grasp Quality Deep Neural Networks for Robotic Object Grasping**, Imperial College London, Undergraduate Open Day, for an audience of 200+, July 2019 and for the Industry Partners Open Day, July 2019

TECHNICAL SKILLS

Programming Languages: Python, Hack, C++, C, JavaScript, Go, Haskell, SQL
Deep Learning Frameworks: JAX, PyTorch, TensorFlow, Keras
Libraries & Tools: NumPy, Pandas, Scikit-learn, Git, Docker

SCHOLARSHIPS AND AWARDS

- **Best Academic Paper Award at ICAIF'23:** for the paper JAX-LOB: A GPU-Accelerated limit order book simulator to unlock large scale reinforcement learning for trading
- **EPSRC DTP Scholarship:** PhD funding for 4 years
- **Corporate Partnership Programme Award:** Outstanding Master's Thesis
- **Distinguished Status Award:** Outstanding Master's Thesis
- **HackCambridge MLH Hackathon:** 2nd place overall (100+ teams)
- **Imperial College Hackathon:** 1st place, Best WebApp (100+ teams) 2nd place, Best Game (100+ teams)
- **Department of Computing Corporate Partnership Program:** Outstanding 1st year project